Anatomy of Flowering Plants

Comprehensive Study Material on Plant Tissue Systems and Internal Structure of Dicotyledonous and Monocotyledonous Plants

1. Learning Objectives & Chapter Overview

When observing larger living organisms—both plants and animals—one can easily notice structural similarities and variations in their external morphology. Similarly, examining the internal structure reveals both fundamental similarities and distinct differences. The study of the internal structure and functional organisation of plants is called **anatomy**.

Plants have **cells** as their basic unit. Cells are organised into **tissues**, and in turn, tissues are organised into **organs** (roots, stems, and leaves). Different organs in a plant exhibit marked differences in their internal structure. Furthermore, within angiosperms (flowering plants), **monocotyledons (monocots)** and **dicotyledons (dicots)** are anatomically distinct. Internal structures also display specific adaptations to diverse environments.

Key Competencies for NEET UG

- Classify plant tissues into the three fundamental tissue systems based on their location and structure: **Epidermal**, **Ground (Fundamental)**, and **Vascular (Conducting)** tissue systems.
- Distinguish between bean-shaped and dumb-bell-shaped guard cells and identify the components of the **stomatal apparatus**.
- Differentiate between **radial**, **conjoint closed**, and **conjoint open** vascular bundles and associate them with specific plant organs.
- Compare and contrast the transverse sections (T.S.) of **dicotyledonous** and **monocotyledonous roots, stems, and leaves**.
- Understand the functional significance of specialised structures such as **Casparian strips**, **starch sheath**, **bundle sheath**, and **bulliform cells**.

2. The Tissue System

Plant tissues are broadly classified into **meristematic** (apical, lateral, and intercalary) and **permanent** (simple and complex) tissues based on the types of cells present and their capacity to divide. The main functions of tissues include assimilation of food and its storage, transportation of water, minerals, and photosynthates, and providing mechanical support.

Tissues also vary depending on their location in the plant body. Their structure and function are directly dependent on their position. On the basis of their **structure and location**, there are three types of tissue systems in flowering plants:

1. **The Epidermal Tissue System**
2. **The Ground (or Fundamental) Tissue System**
3. **The Vascular (or Conducting) Tissue System**

2.1 Epidermal Tissue System

The epidermal tissue system forms the outer-most protective covering of the whole primary plant body. It comprises three main components:

- **Epidermal cells**
- **Stomata**

- **Epidermal appendages** (trichomes and hairs)

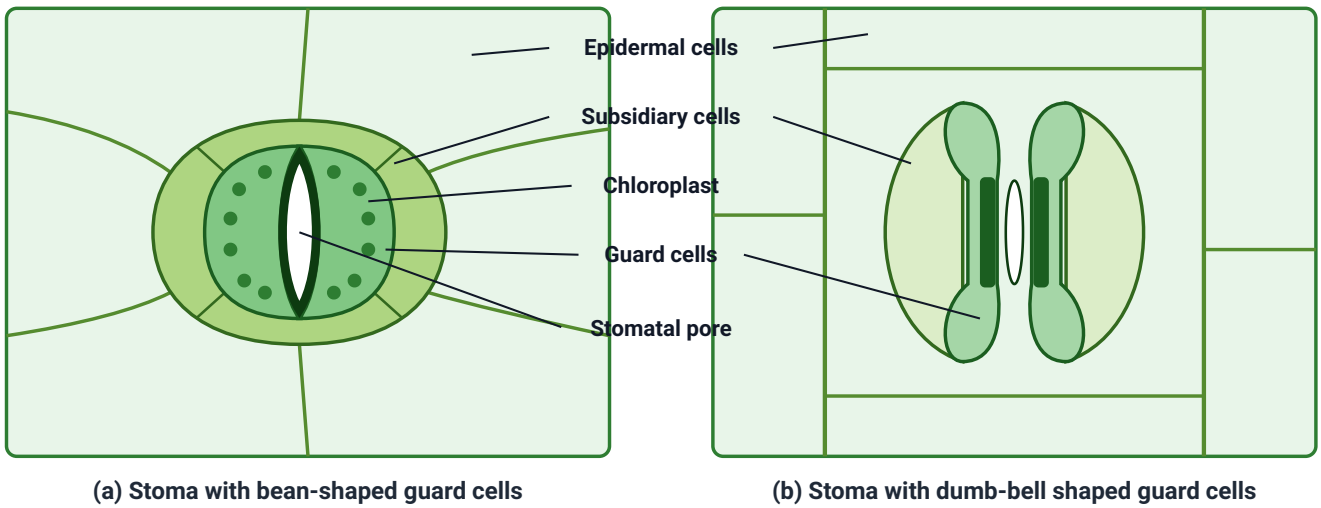
A. Epidermis and Cuticle

- The **epidermis** is the outermost layer of the primary plant body.
- It is made up of elongated, compactly arranged cells (without intercellular spaces) that form a continuous layer.
- The epidermis is usually **single-layered**.
- Epidermal cells are **parenchymatous** in nature, containing a small amount of cytoplasm lining the cell wall and a large central vacuole.
- The outside of the epidermis is often covered with a waxy thick layer called the **cuticle**, which prevents the loss of water. **Cuticle is absent in roots.**

B. Stomata and the Stomatal Apparatus

- **Stomata** are minute structures present in the epidermis of leaves (and young green stems). Stomata regulate the process of **transpiration** and **gaseous exchange**.
- Each stoma is composed of two cells known as **guard cells**, which enclose a **stomatal pore (aperture)**.
- **Shape of Guard Cells:**
  - In dicotyledonous plants, the guard cells are **bean-shaped** (or kidney-shaped).
  - In grasses (monocotyledons), the guard cells are **dumb-bell shaped**.
- **Wall Thickness of Guard Cells:** The outer walls of guard cells (away from the stomatal pore) are **thin**, whereas the inner walls (towards the stomatal pore) are **highly thickened**.
- Guard cells possess **chloroplasts** and regulate the opening and closing of stomata.
- Sometimes, a few epidermal cells in the vicinity of the guard cells become specialised in their shape and size; these are known as **subsidiary cells**.
- The **stomatal aperture**, **guard cells**, and the surrounding **subsidiary cells** are together called the **stomatal apparatus**:

$$\text{Stomatal Apparatus} = \text{Stomatal Pore (Aperture)} + \text{Guard Cells} + \text{Subsidiary Cells}$$



Tutorial Diagram 1: Structure of the Stomatal Apparatus — (a) Dicotyledonous stoma with bean-shaped guard cells; (b) Grass (monocot) stoma with dumb-bell-shaped guard cells.

C. Epidermal Appendages: Root Hairs and Trichomes

The cells of the epidermis bear a number of hairs, which differ significantly between roots and stems:

- **Root Hairs:** They are **unicellular elongations** of the epidermal (epiblema) cells. They help absorb water and minerals from the soil by increasing surface area.
- **Trichomes:** On the stem, the epidermal hairs are called trichomes.
  - Trichomes in the shoot system are usually **multicellular**.
  - They may be **branched or unbranched**, and **soft or stiff**.
  - They may even be **secretory**.
  - **Function:** Trichomes help in preventing water loss due to transpiration.

2.2 The Ground (or Fundamental) Tissue System

- **Definition:** All tissues **except epidermis and vascular bundles** constitute the **ground tissue**. It forms the main bulk of the plant body.
- **Cellular Composition:** It consists of simple permanent tissues such as **parenchyma**, **collenchyma**, and **sclerenchyma**.
- **Location in Stems and Roots:** Ground tissue is divided into three main zones—**cortex**, **pericycle**, and **pith**. Parenchymatous cells are usually present in the cortex, pericycle, pith, and **medullary rays** in the primary stems and roots.
- **Location in Leaves:** In leaves, the ground tissue consists of thin-walled chloroplast-containing parenchymatous cells and is called **mesophyll**.

2.3 The Vascular (or Conducting) Tissue System

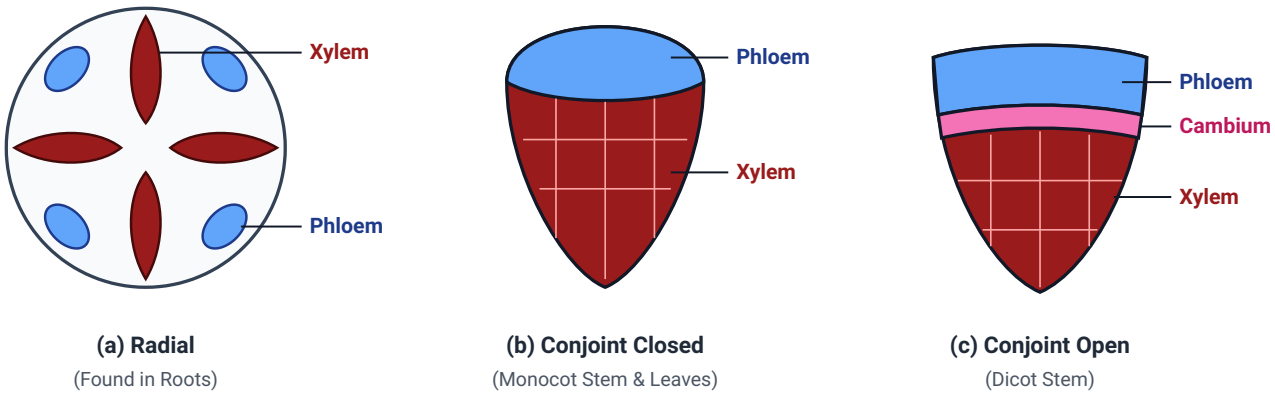
The vascular system consists of complex permanent tissues: the **phloem** and the **xylem**. The xylem and phloem together constitute **vascular bundles**, which form the conducting tissue and translocate water, minerals, and food material.

A. Classification Based on the Presence or Absence of Cambium

- **Open Vascular Bundles:** In dicotyledonous stems, a strip of meristematic tissue called **cambium** (specifically intrafascicular cambium) is present between phloem and xylem. Because of the presence of cambium, such vascular bundles possess the ability to form secondary xylem and phloem tissues, and hence are called **open vascular bundles**.
- **Closed Vascular Bundles:** In monocotyledons, the vascular bundles have **no cambium** present in them. Since they do not form secondary tissues, they are referred to as **closed vascular bundles**.

B. Classification Based on Relative Arrangement of Xylem and Phloem

- **Radial Vascular Bundles:** When xylem and phloem within a vascular bundle are arranged in an **alternate manner along different radii**, the arrangement is called **radial**. Radial vascular bundles are characteristic of **roots** (both dicot and monocot roots).
- **Conjoint Vascular Bundles:** When xylem and phloem are **jointly situated along the same radius** of vascular bundles, the arrangement is called **conjoint**. Such vascular bundles are common in **stems and leaves**. Conjoint vascular bundles usually have the **phloem located only on the outer side of the xylem**.



Tutorial Diagram 2: Types of Vascular Bundles — (a) Radial arrangement along different radii; (b) Conjoint closed without cambium; (c) Conjoint open with cambium strip between phloem and xylem.

3. Anatomy of Dicotyledonous and Monocotyledonous Plants

To understand the tissue organisation of roots, stems, and leaves, we study the transverse sections (T.S.) or vertical sections (V.S.) of the mature zones of these organs.

3.1 Dicotyledonous Root (Example: Sunflower Root)

A transverse section of a primary dicotyledonous root (such as sunflower) shows the following internal tissue organisation from the periphery towards the centre:

1. **Epiblema (Epidermis):** It is the outermost single layer. Cuticle and stomata are absent. Many of the cells of the epiblema protrude outward in the form of **unicellular root hairs**.
2. **Cortex:** It consists of several layers of thin-walled **parenchyma cells with intercellular spaces**.
3. **Endodermis:** It is the innermost layer of the cortex. It comprises a **single layer of barrel-shaped cells without any intercellular spaces**. The **tangential as well as radial walls** of the endodermal cells have a deposition of a water-impermeable, waxy material called **suberin** in the form of **Casparian strips**.
4. **Pericycle:** Next to the endodermis lies a few layers of thick-walled parenchymatous cells referred to as the pericycle. **Initiation of lateral roots** and **vascular cambium during secondary growth** takes place in these cells.
5. **Vascular Bundles:** Vascular bundles are **radial** and **exarch** (protoxylem towards the periphery, metaxylem towards the centre). There are usually **two to four (diarch to tetrarch)** xylem and phloem patches. Later, a **cambium ring** develops between the xylem and phloem for secondary growth.
6. **Conjunctive Tissue:** The **parenchymatous cells** which lie between the xylem and the phloem are called conjunctive tissue.
7. **Pith:** The central pith is **small or inconspicuous**.
8. **Stele:** All tissues on the **inner side of the endodermis**—namely the pericycle, vascular bundles, and pith—together constitute the **stele**:

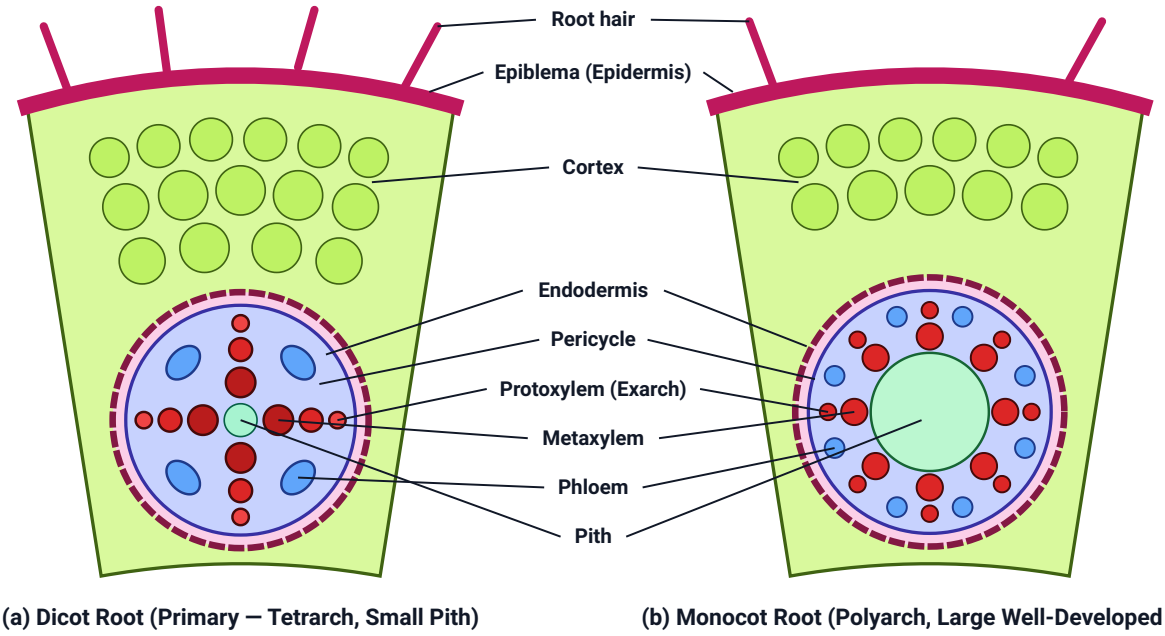
$$\text{Stele} = \text{Pericycle} + \text{Vascular Bundles} + \text{Pith (and Conjunctive Tissue)}$$

3.2 Monocotyledonous Root

The anatomy of the monocot root is similar to the dicot root in many respects: it has an epidermis (epiblema with unicellular root hairs), cortex, endodermis (with Casparian strips), pericycle, radial vascular bundles, and pith. However, it exhibits three major diagnostic differences:

1. **Number of Xylem Bundles (Polyarch Condition):** Compared to the dicot root which has fewer (2 to 4) xylem bundles, there are usually **more than six (polyarch)** xylem bundles in the monocot root.
2. **Pith:** The pith is **large and well-developed** (parenchymatous).

3. **Secondary Growth:** Monocotyledonous roots **do not undergo any secondary growth** (vascular cambium does not form).



Tutorial Diagram 3: Transverse Sections (T.S.) of (a) Dicotyledonous Root and (b) Monocotyledonous Root showing radial exarch vascular bundles and differences in xylem bundle number and pith size.

3.3 Dicotyledonous Stem

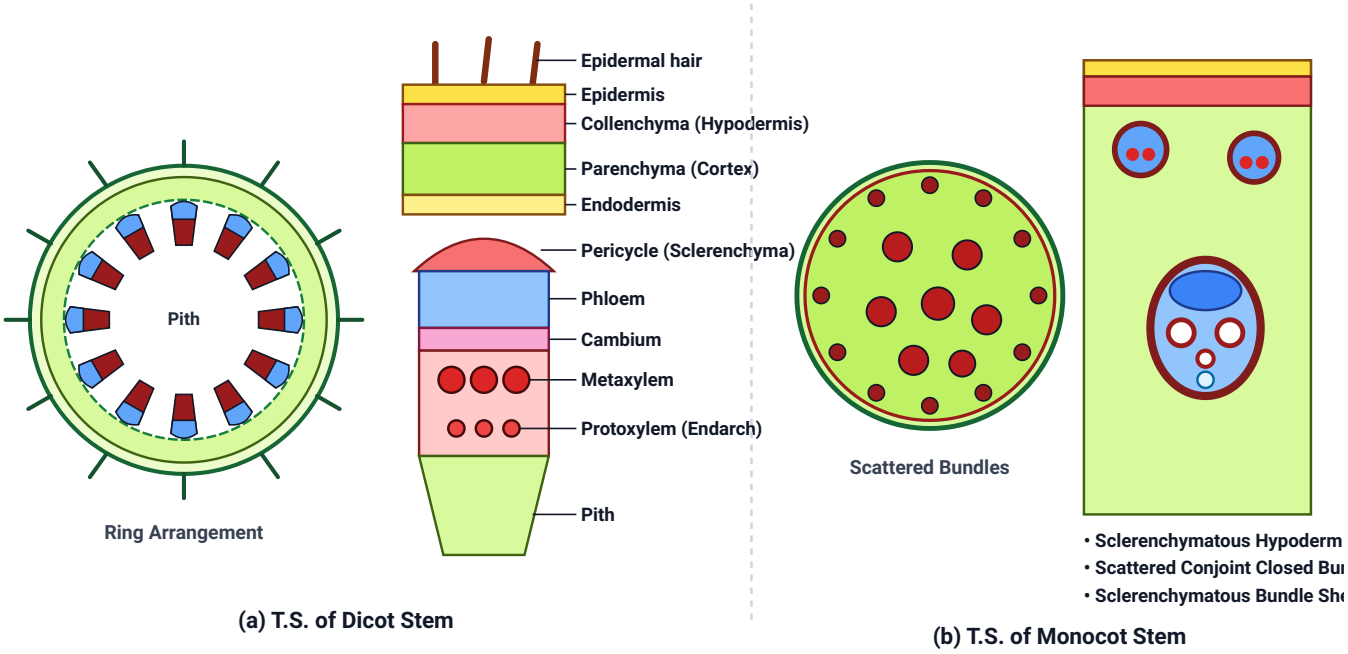
The transverse section of a typical young dicotyledonous stem shows the following regions from the outside inward:

1. **Epidermis:** It is the outermost protective layer of the stem. It is covered with a thin layer of **cuticle**, and it may bear multicellular **trichomes** and a few **stomata**.
2. **Cortex:** The cells arranged in multiple layers between the epidermis and pericycle constitute the cortex. It consists of **three sub-zones**:
   - **Hypodermis (Outer sub-zone):** Consists of a few layers of **collenchymatous cells** just below the epidermis, which provide **mechanical strength** to the young stem.
   - **Cortical Layers (Middle sub-zone):** Below the hypodermis, consisting of rounded, thin-walled **parenchymatous cells** with conspicuous intercellular spaces.
   - **Endodermis (Innermost sub-zone):** The innermost layer of the cortex. The cells of the endodermis are rich in **starch grains**, and the layer is therefore also referred to as the **starch sheath**.
3. **Pericycle:** Present on the inner side of the endodermis and above the phloem in the form of **semi-lunar (half-moon shaped) patches of sclerenchyma**.
4. **Medullary Rays:** In between the vascular bundles, there are a few layers of radially placed parenchymatous cells, which constitute the medullary rays.
5. **Vascular Bundles:** A large number of vascular bundles are arranged in a **ring**; the **'ring' arrangement of vascular bundles is a characteristic feature of the dicot stem**. Each vascular bundle is **conjoint**, **open** (possesses cambium), and with **endarch protoxylem** (protoxylem towards the centre/pith, metaxylem towards the periphery).
6. **Pith (Medulla):** A large number of rounded, **parenchymatous cells** with large intercellular spaces that occupy the central portion of the stem constitute the pith.

3.4 Monocotyledonous Stem

The transverse section of a monocotyledonous stem shows the following characteristic anatomical features:

1. **Epidermis:** Outermost protective layer covered with cuticle; epidermal hairs (trichomes) are absent.
2. **Hypodermis:** The monocot stem has a **sclerenchymatous hypodermis** (2–3 layers of sclerenchyma just below the epidermis providing mechanical rigidity).
3. **Ground Tissue:** There is a **large, conspicuous parenchymatous ground tissue** extending from below the hypodermis to the centre. It is **not differentiated** into cortex, endodermis, pericycle, pith, or medullary rays.
4. **Vascular Bundles:**
   - A large number of vascular bundles are **scattered** throughout the ground tissue.
   - Each vascular bundle is surrounded by a **sclerenchymatous bundle sheath**.
   - Vascular bundles are **conjoint and closed** (cambium is absent) with **endarch protoxylem**.
   - **Size Gradient: Peripheral vascular bundles are generally smaller** than the centrally located ones.
   - **Special Features: Phloem parenchyma is absent**, and **water-containing cavities** are present within the vascular bundles.



Tutorial Diagram 4: Transverse Sections (T.S.) of (a) Dicot Stem (showing ring arrangement of conjoint open vascular bundles and differentiated cortex) and (b) Monocot Stem (showing scattered conjoint closed vascular bundles in undifferentiated ground tissue).

3.5 Dorsiventral (Dicotyledonous) Leaf

The vertical section (V.S.) of a dorsiventral leaf through the lamina shows three main parts: **epidermis**, **mesophyll**, and **vascular system**.

1. **Epidermis:**
   - Covers both the **upper surface (adaxial epidermis)** and the **lower surface (abaxial epidermis)** of the leaf.
   - It has a conspicuous **cuticle**.
   - **Stomatal Distribution:** The **abaxial (lower) epidermis generally bears more stomata than the adaxial (upper) epidermis**. The adaxial epidermis may even lack stomata altogether.

2. **Mesophyll:**
   - The tissue between the upper and the lower epidermis is called the mesophyll.
   - Mesophyll possesses **chloroplasts**, carries out photosynthesis, and is made up of **parenchyma**.
   - In a dorsiventral leaf, it is differentiated into **two types of cells**:

    1. **Palisade Parenchyma:** Placed **adaxially** (towards the upper surface); made up of **elongated cells** arranged **vertically and parallel** to each other.
    2. **Spongy Parenchyma:** Oval or round and **loosely arranged** cells situated below the palisade cells and extending to the lower (abaxial) epidermis. There are **numerous large intercellular spaces and air cavities** between these cells (including sub-stomatal cavities).

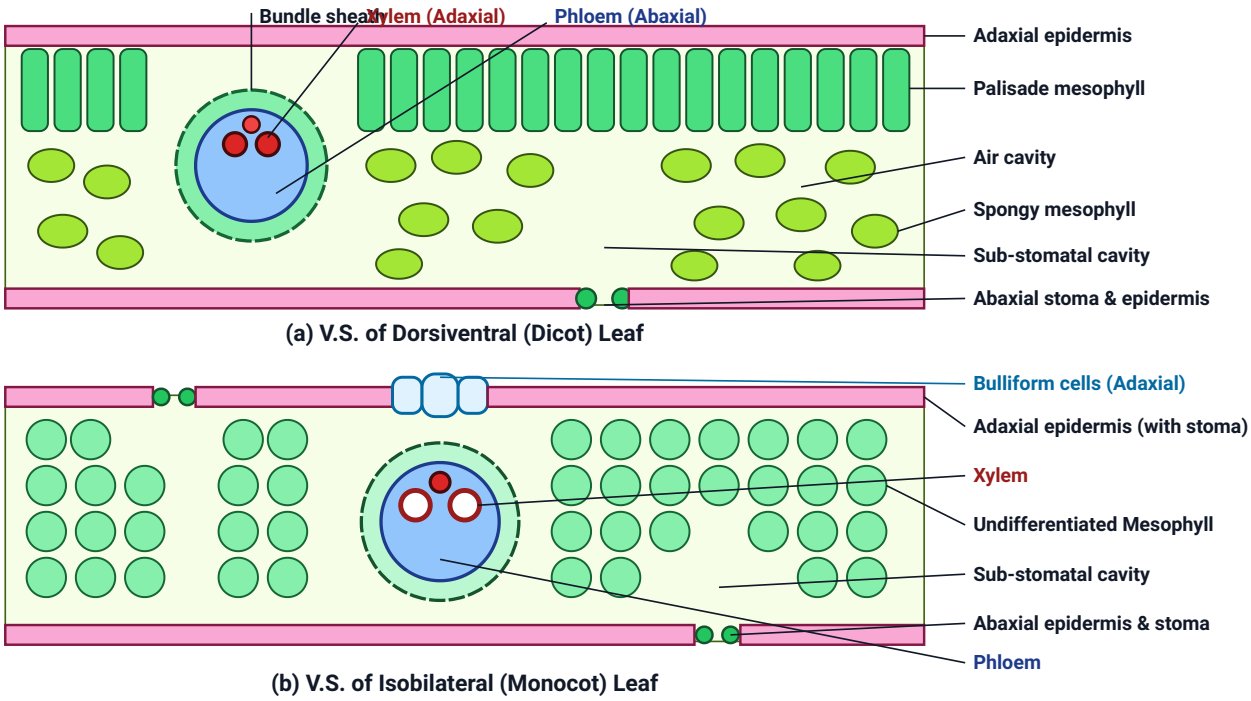
3. **Vascular System:**
    - Includes vascular bundles seen in the **veins and the midrib**. Vascular bundles are **conjoint and closed**.
    - Within the leaf vascular bundle, **xylem is positioned towards the adaxial (upper) epidermis** and **phloem towards the abaxial (lower) epidermis**.
    - **Size of Vascular Bundles:** Dependent on the size of the veins. Because veins vary in thickness in the **reticulate venation** of dicot leaves, vascular bundles of varying sizes are seen in vertical section.
    - The vascular bundles are surrounded by a layer of thick-walled **bundle sheath cells**.

3.6 Isobilateral (Monocotyledonous) Leaf

The anatomy of an isobilateral leaf is similar to that of the dorsiventral leaf in many ways, but it exhibits the following characteristic differences:

1. **Stomatal Distribution:** Stomata are present on **both the surfaces of the epidermis** (both adaxial and abaxial surfaces bear stomata equally).
2. **Undifferentiated Mesophyll:** The mesophyll is **not differentiated** into palisade and spongy parenchyma; it consists of uniform isodiametric photosynthetic parenchyma cells.
3. **Bulliform Cells in Grasses:**
    - In grasses, certain **adaxial epidermal cells along the veins** modify themselves into **large, empty, colourless cells** called **bulliform cells**.
    - **When Turgid:** When the bulliform cells in the leaves have absorbed water and are **turgid**, the **leaf surface is exposed** (uncurled).
    - **When Flaccid:** When they become **flaccid due to water stress**, they make the **leaves curl inwards to minimise water loss** (reduce transpiration).

4. **Vascular Bundle Sizes:** The **parallel venation** in monocot leaves is reflected in the **near similar sizes of vascular bundles (except in main veins)** as seen in vertical sections of the leaves.



Tutorial Diagram 5: Vertical Sections (V.S.) of (a) Dorsiventral (Dicot) Leaf showing palisade and spongy mesophyll differentiation, and (b) Isobilateral (Monocot) Leaf showing undifferentiated mesophyll, stomata on both surfaces, and bulliform cells.

4. Comparative Summary Tables & Key Anatomical Results

In competitive examinations like NEET UG, direct comparative questions between dicot and monocot organs are among the highest-yield topics. Review the three master comparison tables below carefully.

4.1 Comparison: Dicotyledonous Root vs. Monocotyledonous Root

Anatomical Feature	Dicotyledonous Root	Monocotyledonous Root
Pericycle Function	Gives rise to lateral roots and vascular cambium during secondary growth.	Gives rise to lateral roots only (no secondary growth).
Number of Xylem/Phloem Bundles	Usually **2 to 4** (diarch to tetrarch) xylem and phloem patches.	Usually **more than six (polyarch)** xylem bundles.
Pith	**Small or inconspicuous**.	**Large and well-developed**.
Cambium & Secondary Growth	A cambium ring develops later between xylem and phloem; **undergoes secondary growth**.	Cambium is absent; **does not undergo any secondary growth**.

4.2 Comparison: Dicotyledonous Stem vs. Monocotyledonous Stem

Anatomical Feature	Dicotyledonous Stem	Monocotyledonous Stem
Trichomes (Epidermal Hairs)	Usually present (multicellular).	Absent.
Hypodermis	**Collenchymatous** (provides mechanical strength to young stem).	**Sclerenchymatous**.
Ground Tissue Differentiation	Differentiated into cortex (hypodermis, general cortex, endodermis/starch sheath), pericycle, medullary rays, and central pith.	**Not differentiated**; forms a large, conspicuous parenchymatous ground tissue.
Pericycle	Present above phloem as **semi-lunar patches of sclerenchyma**.	Absent (instead, each vascular bundle is surrounded by a **sclerenchymatous bundle sheath**).
Arrangement of Vascular Bundles	Arranged in a **ring** (characteristic of dicot stem).	**Scattered** throughout the ground tissue (peripheral ones smaller, central ones larger).
Type of Vascular Bundles	**Conjoint, open** (cambium present), with endarch protoxylem.	**Conjoint, closed** (cambium absent), with endarch protoxylem.
Phloem Parenchyma & Water Cavities	Phloem parenchyma is present; water-containing cavities are absent.	**Phloem parenchyma is absent**; **water-containing cavities are present** within vascular bundles.

4.3 Comparison: Dorsiventral (Dicot) Leaf vs. Isobilateral (Monocot) Leaf

Anatomical Feature	Dorsiventral (Dicotyledonous) Leaf	Isobilateral (Monocotyledonous) Leaf
Stomatal Distribution	**Abaxial (lower) epidermis generally bears more stomata** than adaxial (upper) epidermis; adaxial may even lack stomata.	Stomata are present on **both the surfaces** (adaxial and abaxial) of the epidermis.
Guard Cell Shape	**Bean-shaped** (kidney-shaped).	**Dumb-bell shaped** (in grasses).
Mesophyll	Differentiated into upper **palisade parenchyma** and lower **spongy parenchyma**.	**Not differentiated** into palisade and spongy parenchyma.
Bulliform Cells	Absent.	Present in grasses (large, empty, colourless modified **adaxial epidermal cells** along veins).
Venation & Size of Vascular Bundles	**Reticulate venation**; vascular bundles **vary in size** depending on vein thickness.	**Parallel venation**; vascular bundles are of **near similar sizes** (except in main veins).

5. Diagnostic Identification Key for Unknown Plant Specimens

When observing a microscopic transverse section of an unknown plant material in the laboratory or solving an identification problem in an examination, follow this systematic two-step diagnostic procedure:

Step 1: Determine Whether the Organ is a Root, Stem, or Leaf

- **Look at the Vascular Bundles:**
  - If xylem and phloem are on **different radii (Radial)** and protoxylem is towards the periphery (**Exarch**), plus cuticle is absent and unicellular root hairs are present → **It is a ROOT**.
  - If xylem and phloem are on the **same radius (Conjoint)** and protoxylem is towards the centre (**Endarch**), plus cuticle is present → **It is a STEM**.
  - If the section shows an upper (adaxial) and lower (abaxial) epidermis enclosing chloroplast-rich **mesophyll** and conjoint closed bundles surrounded by bundle sheaths → **It is a LEAF**.

Step 2: Determine Whether It is Dicotyledonous or Monocotyledonous

- **If it is a Root:**
  - 2 to 4 xylem patches + small/inconspicuous pith → **Dicot Root**.
  - More than 6 (polyarch) xylem patches + large well-developed pith → **Monocot Root**.
- **If it is a Stem:**
  - Vascular bundles in a **ring** + **conjoint open** (cambium present) + collenchymatous hypodermis + starch sheath + semi-lunar sclerenchymatous pericycle → **Dicot Stem**.
  - Vascular bundles **scattered** + **conjoint closed** + sclerenchymatous hypodermis + sclerenchymatous bundle sheath + phloem parenchyma absent + water-containing cavities → **Monocot Stem**.
- **If it is a Leaf:**
  - Mesophyll differentiated into palisade and spongy parenchyma + more stomata on abaxial surface → **Dorsiventral (Dicot) Leaf**.
  - Mesophyll undifferentiated + stomata on both surfaces + bulliform cells (in grasses) + near-similar-sized vascular bundles → **Isobilateral (Monocot) Leaf**.

6. Worked Examples (Applied Conceptual Exercises)

Worked Example 1: Microscopic Identification of a Garden Stem

Problem Statement: Suppose you cut a transverse section of a young stem of a plant from your school garden and observe it under the microscope. How would you ascertain whether it is a monocot stem or a dicot stem? Give clear anatomical reasons.

Solution & Reasoning:

Under the microscope, the young stem can be definitively identified as either a dicot stem or a monocot stem by examining the following key anatomical characters:

1. **Arrangement of Vascular Bundles:** If the vascular bundles are arranged in a distinct '**ring**' around a central pith, it is a **dicot stem**. If the vascular bundles are **scattered** throughout the ground tissue (with smaller bundles towards the periphery and larger ones towards the centre), it is a **monocot stem**.
2. **Nature of Vascular Bundles:** In a **dicot stem**, vascular bundles are **conjoint and open** (a strip of cambium is present between phloem and xylem). In a **monocot stem**, vascular bundles are **conjoint and closed** (cambium is absent), and each bundle is enclosed by a **sclerenchymatous bundle sheath**.
3. **Hypodermis and Ground Tissue Differentiation:** A **dicot stem** possesses a **collenchymatous hypodermis**, an endodermis rich in starch grains (**starch sheath**), a pericycle made of **semi-lunar patches of sclerenchyma**, medullary rays, and a central pith. Conversely, a **monocot stem** possesses a **sclerenchymatous hypodermis** and an undifferentiated parenchymatous ground tissue.

Worked Example 2: Diagnosing a Plant Organ from Histological Clues

Problem Statement: The transverse section of a plant material shows the following anatomical features: (a) the vascular bundles are conjoint, scattered, and surrounded by sclerenchymatous bundle sheaths, and (b) phloem parenchyma is absent. What will you identify it as?

Solution & Reasoning:

- The presence of **conjoint** vascular bundles indicates that the organ is not a root (since roots have radial bundles).
- The **scattered arrangement** of vascular bundles in the ground tissue, surrounded by **sclerenchymatous bundle sheaths**, along with the **absence of phloem parenchyma** (and presence of water-containing cavities), are hallmark diagnostic features of a **Monocotyledonous Stem (Monocot Stem)**.

Worked Example 3: Tissue Systems and Functional Significance of Plant Anatomy

Problem Statement: (a) Name the three basic tissue systems in flowering plants and list the tissues present under each system. (b) How is the study of plant anatomy useful to us?

Solution & Reasoning:

(a) The Three Basic Tissue Systems:

1. **Epidermal Tissue System:** Comprises epidermal cells (parenchymatous), stomata (guard cells and subsidiary cells), and epidermal appendages (unicellular root hairs and multicellular stem trichomes).
2. **Ground (or Fundamental) Tissue System:** Comprises simple permanent tissues—**parenchyma, collenchyma, and sclerenchyma**—forming the cortex, pericycle, pith, medullary rays (in stems and roots), and mesophyll (in leaves).
3. **Vascular (or Conducting) Tissue System:** Comprises complex permanent tissues—**xylem and phloem**—which form vascular bundles.

(b) Utility of Studying Plant Anatomy:

- It helps us understand the internal structural organisation of different plant organs (roots, stems, leaves) and how structure correlates with physiological functions like water absorption, photosynthesis, transpiration, and translocation.
- It enables accurate taxonomic distinction between **monocotyledons and dicotyledons** even from small fragments of plant material.
- It explains how plants adapt internally to diverse environmental conditions (for example, thick cuticle, trichomes, and bulliform cells in grasses to minimise water loss during water stress).

7. NEET UG Practice Questions with Complete Worked Solutions

Note: All questions below are original exam-aligned practice questions designed strictly in accordance with the NEET UG syllabus and pattern.

Practice Question 1 [Single-Correct MCQ]

Which of the following statements is **incorrect** regarding the epidermal tissue system of flowering plants?

(A) Epidermal cells are parenchymatous with a small amount of cytoplasm lining the cell wall and a large vacuole.

(B) The outer walls of guard cells (away from the stomatal pore) are highly thickened, whereas the inner walls (towards the stomatal pore) are thin.

(C) Cuticle is a waxy thick layer outside the epidermis that prevents loss of water, and it is absent in roots.

(D) Root hairs are unicellular elongations of epidermal cells, whereas trichomes in the shoot system are usually multicellular.

Correct Answer: (B)

Worked Solution: In the stomatal apparatus, each stoma is enclosed by two guard cells. The **outer walls of guard cells (away from the stomatal pore) are thin**, and the **inner walls (towards the stomatal pore) are highly thickened**. Therefore, statement (B) reverses the wall thicknesses and is incorrect. Statements (A), (C), and (D) are completely accurate anatomical facts.

Practice Question 2 [Single-Correct MCQ]

What constitutes the **stomatal apparatus** in the leaf epidermis?

(A) Stomatal aperture and guard cells only

(B) Guard cells, chloroplasts, and bulliform cells

(C) Stomatal aperture, guard cells, and surrounding subsidiary cells

(D) Epidermal cells, cuticle, trichomes, and guard cells

Correct Answer: (C)

Worked Solution: A few epidermal cells in the vicinity of the guard cells become specialised in their shape and size and are known as **subsidiary cells**. The **stomatal aperture, guard cells, and the surrounding subsidiary cells** together constitute the **stomatal apparatus**.

Practice Question 3 [Single-Correct MCQ]

In the primary dicotyledonous root, deposition of water-impermeable, waxy material suberin in the form of **Casparian strips** occurs on:

(A) Only the radial walls of the pericycle cells

(B) Both tangential and radial walls of the endodermal cells

(C) Only the outer tangential walls of the epiblema cells

(D) The inner walls of conjunctive tissue cells

Correct Answer: (B)

Worked Solution: The innermost layer of the cortex in roots is the **endodermis**, which comprises a single layer of barrel-shaped cells without any intercellular spaces. The **tangential as well as radial walls** of the endodermal cells have a deposition of water-impermeable, waxy material **suberin** in the form of **Casparian strips**.

Practice Question 4 [Single-Correct MCQ]

During secondary growth in a dicotyledonous root, initiation of lateral roots and vascular cambium takes place in the cells of the:

(A) Endodermis

(B) Pericycle

(C) Epiblema

(D) Medullary rays

Correct Answer: (B)

Worked Solution: Next to the endodermis in a dicot root lies a few layers of thick-walled parenchymatous cells referred to as the **pericycle**. Initiation of **lateral roots** and **vascular cambium** during secondary growth takes place in these pericycle cells.

Practice Question 5 [Single-Correct MCQ]

Which of the following combinations of tissues constitutes the **stele** in a root?

(A) Epiblema, cortex, and endodermis

(B) Endodermis, pericycle, and cortex

(C) Pericycle, vascular bundles, and pith

(D) Xylem, phloem, and endodermis only

Correct Answer: (C)

Worked Solution: By definition, all tissues on the **inner side of the endodermis**—such as the **pericycle, vascular bundles, and pith**—constitute the **stele**. Note that the endodermis itself is the innermost layer of the cortex and is outside the stele.

Practice Question 6 [Multiple-Statement Evaluation]

Read the following statements regarding the transverse section of a young **dicotyledonous stem**:

1. The hypodermis consists of a few layers of collenchymatous cells just below the epidermis.
2. The cells of the endodermis are rich in starch grains and the layer is referred to as the starch sheath.
3. The pericycle is present on the inner side of the endodermis and above the phloem in the form of semi-lunar patches of parenchyma.
4. Vascular bundles are arranged in a ring, and each bundle is conjoint, open, with endarch protoxylem.

Which of the above statements are **correct**?

(A) 1, 2, and 4 only

(B) 1, 2, and 3 only

(C) 2, 3, and 4 only

(D) 1, 2, 3, and 4

Correct Answer: (A)

Worked Solution: Statements 1, 2, and 4 are correct. Statement 3 is incorrect because the pericycle in a dicot stem is present above the phloem in the form of semi-lunar patches of **sclerenchyma** (not parenchyma).

Practice Question 7 [Multiple-Statement Evaluation]

Consider the following features concerning the anatomy of a **monocotyledonous stem**:

1. It possesses a sclerenchymatous hypodermis.
2. Peripheral vascular bundles are generally larger than the centrally located ones.
3. Each vascular bundle is surrounded by a sclerenchymatous bundle sheath.
4. Phloem parenchyma is absent, and water-containing cavities are present within the vascular bundles.

Select the option that includes **only the correct statements**:

(A) 1, 2, and 3

(B) 1, 3, and 4

(C) 2, 3, and 4

(D) 1 and 4 only

Correct Answer: (B)

Worked Solution: Statements 1, 3, and 4 are correct. Statement 2 is incorrect because in a monocot stem, **peripheral vascular bundles are generally smaller** than the centrally located ones.

Practice Question 8 [Single-Correct MCQ]

In grasses, certain cells in the leaf epidermis modify themselves into large, empty, colourless cells that help minimise water loss by curling the leaf inwards during water stress. Where are these cells located and what are they called?

(A) Abaxial epidermal cells along the veins; called subsidiary cells

(B) Adaxial epidermal cells along the veins; called bulliform cells

(C) Mesophyll cells between veins; called spongy parenchyma

(D) Bundle sheath cells around main veins; called conjunctive cells

Correct Answer: (B)

Worked Solution: In grasses (isobilateral monocot leaves), certain **adaxial (upper) epidermal cells along the veins** modify themselves into large, empty, colourless cells called **bulliform cells**. When turgid, they expose the leaf surface; when flaccid due to water stress, they make the leaves curl inwards to minimise water loss.

Practice Question 9 [Assertion–Reason]

Assertion (A): In a vertical section of an isobilateral (monocotyledonous) leaf, the vascular bundles are seen to be of near similar sizes (except in main veins).

Reason (R): Monocotyledonous leaves possess parallel venation in which veins run parallel and are of nearly equal thickness.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Solution: The size of vascular bundles in a leaf is directly dependent on the size (thickness) of the veins. In dicot leaves, reticulate venation has veins of varying thickness, producing vascular bundles of varying sizes. In contrast, the **parallel venation in monocot leaves** is reflected in the **near similar sizes of vascular bundles** (except in main veins) as seen in vertical sections. Thus, both (A) and (R) are true and (R) correctly explains (A).

Practice Question 10 [Assertion–Reason]

Assertion (A): Vascular bundles in monocotyledonous stems are referred to as closed vascular bundles.

Reason (R): In monocotyledonous stems, each vascular bundle is completely surrounded by a sclerenchymatous bundle sheath.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (B)

Worked Solution: Assertion (A) is true: vascular bundles in monocots are called **closed**. Reason (R) is also true: each vascular bundle in a monocot stem is surrounded by a sclerenchymatous bundle sheath. However, vascular bundles are called "closed" **because they have no cambium present in them and do not form secondary tissues**—not because of the bundle sheath. Hence, both statements are true, but (R) is not the correct explanation of (A).

Practice Question 11 [Matching Type]

Match the anatomical structures in **Column I** with their characteristic locations or descriptions in **Column II**:

Column I (Structure)	Column II (Location / Characteristic)
(P) Conjunctive tissue	(i) Endodermis of dicot stem rich in starch grains
(Q) Starch sheath	(ii) Parenchymatous cells lying between xylem and phloem in roots
(R) Medullary rays	(iii) Dumb-bell shaped in grasses
(S) Guard cells	(iv) Radially placed parenchymatous cells between vascular bundles in dicot stem

(A) P–(ii), Q–(i), R–(iv), S–(iii)

(B) P–(iv), Q–(i), R–(ii), S–(iii)

(C) P–(ii), Q–(iii), R–(iv), S–(i)

(D) P–(i), Q–(ii), R–(iii), S–(iv)

Correct Answer: (A)

Worked Solution:
• **(P) Conjunctive tissue:** Parenchymatous cells which lie between the xylem and the phloem in roots → (ii).
• **(Q) Starch sheath:** Endodermis of dicot stem whose cells are rich in starch grains → (i).
• **(R) Medullary rays:** A few layers of radially placed parenchymatous cells between vascular bundles in a dicot stem → (iv).
• **(S) Guard cells:** Bean-shaped in dicots and dumb-bell shaped in grasses → (iii).

Practice Question 12 [Single-Correct MCQ]

Compare the vertical section of a dorsiventral (dicot) leaf with that of an isobilateral (monocot) leaf and identify the **correct** statement:

(A) In a dorsiventral leaf, the spongy parenchyma is placed adaxially and the palisade parenchyma is placed abaxially.

(B) In a dorsiventral leaf, the adaxial epidermis generally bears more stomata than the abaxial epidermis.

(C) In an isobilateral leaf, the mesophyll is not differentiated into palisade and spongy parenchyma.

(D) Vascular bundles in leaf veins are radial and open.

Correct Answer: (C)

Worked Solution: In an isobilateral (monocot) leaf, the mesophyll is **not differentiated into palisade and spongy parenchyma**, making (C) correct. In a dorsiventral leaf, palisade parenchyma is placed adaxially (upper side) and spongy parenchyma is below it extending to the lower (abaxial) epidermis, and the abaxial epidermis bears more stomata than the adaxial epidermis. Leaf vascular bundles are conjoint and closed.

8. Common Pitfalls & High-Yield NEET Revision Traps

Watch Out for These Frequent Exam Traps!

1. **Root Hairs vs. Stem Trichomes (Cellularity Trap):** Students frequently confuse the cellularity of epidermal appendages. Remember: **Root hairs are always UNICELULLAR** elongations of epiblema cells, whereas **stem trichomes are usually MULTICELLULAR**.
2. **Guard Cell Wall Thickness Trap:** Do not mix up inner and outer walls! In guard cells, the **outer wall (away from pore) is THIN** and the **inner wall (towards pore) is HIGHLY THICKENED**. Also remember that guard cells **possess chloroplasts**, unlike ordinary epidermal cells.
3. **Hypodermis Composition (Dicot vs. Monocot Stem):**
   - Dicot Stem Hypodermis = **Collenchymatous**.
   - Monocot Stem Hypodermis = **Sclerenchymatous**.
4. **Pericycle Composition (Dicot Root vs. Dicot Stem):**
   - Dicot Root Pericycle = **Thick-walled parenchymatous cells** (gives rise to lateral roots and vascular cambium).
   - Dicot Stem Pericycle = **Semi-lunar patches of sclerenchyma** above the phloem.
5. **Endodermis in Roots vs. Dicot Stems:**
   - In Roots: Endodermis has **Casparian strips (suberin)** on both tangential and radial walls.
   - In Dicot Stems: Endodermis is rich in starch grains and is called the **starch sheath**.
6. **Adaxial vs. Abaxial Leaf Surfaces:**
   - **Adaxial** = Upper surface (faces towards the stem axis). Contains palisade parenchyma underneath in dicots, and **bulliform cells** in grasses.
   - **Abaxial** = Lower surface (faces away from the stem axis). Bears **more stomata** in dorsiventral (dicot) leaves.

7. **Size of Vascular Bundles in Monocot Stems:** Peripheral vascular bundles are **smaller**; centrally located vascular bundles are **larger**.

9. Concise Final Recap

- **Three Tissue Systems:**
  1. *Epidermal:* Epidermis, cuticle (absent in roots), stomata (bean-shaped guard cells in dicots; dumb-bell-shaped in grasses), unicellular root hairs, multicellular trichomes.
  2. *Ground:* All tissues except epidermis and vascular bundles; comprises parenchyma, collenchyma, and sclerenchyma (cortex, pericycle, pith, medullary rays, and leaf mesophyll).
  3. *Vascular:* Xylem and phloem forming vascular bundles—**Radial** (in roots) vs. **Conjoint** (in stems and leaves); **Open** (with cambium, in dicot stems) vs. **Closed** (without cambium, in monocots and leaves).
- **Dicot vs. Monocot Root:** Both have radial exarch bundles, Casparian strips on endodermis, and pericycle. Dicot root has **2–4 xylem patches**, **small pith**, and shows secondary growth. Monocot root has **more than 6 (polyarch) xylem bundles**, **large pith**, and no secondary growth.
- **Dicot vs. Monocot Stem:** Dicot stem has a **collenchymatous hypodermis**, **starch sheath**, **semi-lunar sclerenchymatous pericycle**, medullary rays, pith, and **conjoint open bundles in a ring**. Monocot stem has a **sclerenchymatous hypodermis**, undifferentiated ground tissue, and **scattered conjoint closed bundles** with **sclerenchymatous bundle sheaths**, **no phloem parenchyma**, and **water-containing cavities**.
- **Dicot (Dorsiventral) vs. Monocot (Isobilateral) Leaf:** Dicot leaf has more stomata on the abaxial surface, mesophyll differentiated into **palisade and spongy parenchyma**, and variably sized vascular bundles (reticulate venation). Monocot leaf has stomata on both surfaces, **undifferentiated mesophyll**, **bulliform cells** on the adaxial epidermis in grasses, and nearly equal-sized vascular bundles (parallel venation).